Assigment 2 - Prompting Techniques in Health Services

Start Assignment

- Due Saturday by 11:59pm
- Points 50
- Submitting a file upload

Prompting Techniques in Health Services

Individual paper presentation | Weeks 5- 9 | 50 points

Present your assigned paper in plain language, focusing on how the prompting method works and how it could support a healthcare task.

Course coverage: Week 5: healthcare text; Week 6: prompting and AI systems; Week 7: quality and reliability; Week 8: real-world value and monitoring; Week 9: responsible AI.

Student paper allocation

Selected prompting and reasoning papers. Click a title or **Record** for the official publication; **arXiv** opens an alternative author-posted copy.

Student	Assigned paper / authors	Ven
1. Alaa Abukhazneh	Graph of Thoughts: Solving Elaborate Problems with Large Language Models (https://ojs.aaai.org/index.php/AAAI/article/view/29720) — Besta et al.	AAAI 2024 Record (https://ojs.aaai.org/index.php/AAAI (https://arxiv.org/abs/2308.09687)
2. Kim Hagan	Chain of Agents: Large Language Models Collaborating on Long-Context Tasks (https://proceedings.neurips.cc/paper_files/paper/2024/hash/ee71a4b14ec26710b39ee6be113d7750-Abstract-Conference.html) — Zhang et al.	NeurIPS 2024 Record (https://proceedings.neurips.cc/paper_files/pap Abstract-Conference.html)
3. Brian Harmon	Everything of Thoughts: Defying the Law of Penrose Triangle for Thought Generation (https://aclanthology.org/2024.findings-acl.95/) — Ding et al.	Findings of ACL 2024 Record (https://aclanthology.org/2024.findir
4. Kayla Hicks	Mutual Reasoning Makes Smaller LLMs Stronger Problem-Solver (https://proceedings.iclr.cc/paper_files/paper/2025/hash/35514d533cdc278a7780daf0dbe7d0b7-Abstract-Conference.html) — Qi et al. (rStar)	ICLR 2025 Record (https://proceedings.iclr.cc/paper_files/paper/2(Abstract-Conference.html) | arXiv (https://ar
5. Alex Hillman	Skeleton-of-Thought: Prompting LLMs for Efficient Parallel Generation (https://proceedings.iclr.cc/paper_files/paper/2024/hash/03d7e13f0092405804f3a381ade8f3f0-Abstract-Conference.html) — Ning et al.	ICLR 2024 Record (https://proceedings.iclr.cc/paper_files/paper/2(Abstract-Conference.html)
6. Cordel Hooks	Chain-of-Thought Prompting Elicits Reasoning in Large Language Models (https://proceedings.neurips.cc/paper/2022/hash/9d5609613524ecf4f15af0f7b31abca4-Abstract-Conference.html) — Wei et al.	NeurIPS 2022 Record (https://proceedings.neurips.cc/pap Abstract-Conference.html)
7. Grant Imbrock (He/His)	Large Language Models are Zero-Shot Reasoners (https://proceedings.neurips.cc/paper_files/paper/2022/hash/8bb0d291acd4acf06ef112099c16f326-Abstract-Conference.html) — Kojima et al.	NeurIPS 2022 Record (https://proceedings.neurips.cc/paper_files/pap Abstract-Conference.html)
8. Urva LNU (She/Her)	Self-Consistency Improves Chain of Thought Reasoning in Language Models (https://openreview.net/forum?id=1PL1NIMMrw) — Wang et al.	ICLR 2023 Record (https://openreview.net/forum?id=1I (https://arxiv.org/abs/2203.11171)
9. Hathaichanok Poonlert	Least-to-Most Prompting Enables Complex Reasoning in Large Language Models (https://openreview.net/forum?id=WZH7099tgfM) — Zhou et al.	ICLR 2023 Record (https://openreview.net/forum?id=W (https://arxiv.org/abs/2205.10625)

Student	Assigned paper / authors	Ven
10. Domenica Pringle	**Tree of Thoughts: Deliberate Problem Solving with Large Language Models** (https://proceedings.neurips.cc/paper_files/paper/2023/hash/271db9922b8d1f4dd7aaef84ed5ac703-Abstract-Conference.html) — Yao et al.	NeurIPS 2023 **Record** (https://proceedings.neurips.cc/paper_files/pap Abstract-Conference.html)
11. Julia Rehman	**ReAct: Synergizing Reasoning and Acting in Language Models** (https://openreview.net/forum?id=WE_vluYUL-X) — Yao et al.	ICLR 2023 **Record** (https://openreview.net/forum?id=W (https://arxiv.org/abs/2210.03629)
12. Justin Routman	**Self-Refine: Iterative Refinement with Self-Feedback** (https://proceedings.neurips.cc/paper_files/paper/2023/hash/91edff07232fb1b55a505a9e9f6c0ff3-Abstract-Conference.html) — Madaan et al.	NeurIPS 2023 **Record** (https://proceedings.neurips.cc/paper_files/pap Abstract-Conference.html)
13. Saba Tufail	**Reflexion: language agents with verbal reinforcement learning** (https://proceedings.neurips.cc/paper_files/paper/2023/hash/1b44b878bb782e6954cd888628510e90-Abstract-Conference.html) — Shinn et al.	NeurIPS 2023 **Record** (https://proceedings.neurips.cc/paper_files/pap Abstract-Conference.html)
14. Sam Williams	**Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection** (https://proceedings.iclr.cc/paper_files/paper/2024/hash/25f7be9694d7b32d5cc670927b8091e1-Abstract-Conference.html) — Asai et al.	ICLR 2024 **Record** (https://proceedings.iclr.cc/paper_files/paper/2(Abstract-Conference.html)

What your presentation must cover

Presentation Duration: 10 Mins-Presentation + 5 mins QnA

Presentation Venue: In Class (During class hours)

Focus on practical understanding. No coding, formulas, new experiments, or detailed explanations of AI theory are required.

1. **Goal and improvement.** State the problem the paper addresses and what its method changes compared with an earlier approach.
2. **Prompting workflow.** Use a simple flow diagram to show the input, main instructions, steps, and final output. Identify any additional agents, tools, or information sources only when used. Show what the AI is asked to do and the expected answer format. Label simplified or inferred prompts clearly; a short example is encouraged.
3. **Strengths and limitations.** Include at least two strengths and two limitations. Consider errors, time, cost, and ease of use rather than focusing on benchmark tables.
4. **Healthcare application.** Choose one clinical or administrative text task, such as summarization or finding relevant patient records. Identify the text input, expected output, and person who should review it. Distinguish your proposed use from what the paper tested.
5. **Quality and reliability.** Describe how you would check whether answers are correct and complete, try difficult or unfamiliar cases, and decide when an answer needs human review or should not be used. Discuss the paper's evidence in plain language; do not calculate or teach performance measures.
6. **Practical and responsible use.** Discuss likely benefits, effects on staff workload, time or cost, and what should be monitored after use. Address privacy, unfair or unsafe results, and who can correct or stop the system.

Grading:

25 Point Submission + 12.5 Point Presentation + 12.5 Points QnA = 50 Points

Submission: Upload slides before class (PDF or PPT/PPTX), cite the paper, and be ready for questions. Use fictional examples; do not include identifiable patient information.

Hints for your Assignment □

Category	What is assessed
Goal and improvement	Clearly identifies the problem and the method's main improvement.
Technical understanding: prompting workflow	Accurately shows the main steps, instructions, inputs, and outputs in a simple flow diagram.
Strengths and limitations	Provides at least two relevant strengths and two limitations.
Healthcare application	Gives a suitable text-based healthcare task, expected output, and human reviewer.
Quality and reliability	Proposes sensible checks for accuracy, missing information, difficult cases, and when to seek human review.
Real-world usefulness	Considers practical benefit, workload, time or cost, and monitoring.
Responsible use	Identifies privacy, fairness, and safety concerns, with practical safeguards and human oversight.

Assignment-2-Prompting_Techniques.docx (https://uab.instructure.com/courses/1691197/files/83060829?wrap=1) 
(https://uab.instructure.com/courses/1691197/files/83060829/download?download_frd=1) 